

Chapter 4c: Noor Iteration

Fixed Point Theory and Nonlinear Analysis

Based on Pathak's Introduction to Nonlinear Analysis

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Chapter Overview

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- 2 Noor Iteration: Definition and Formulation
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Iteration Methods in Fixed Point Theory

Motivation

Fixed point iteration methods are fundamental algorithms for:

- Finding approximate solutions to nonlinear equations
- Solving problems in numerical analysis
- Convergence analysis of iterative algorithms
- Applications in optimization and differential equations

Progression of Methods

Different iteration methods offer varying levels of complexity and convergence speed:

- 1 **Mann Iteration** (1-step): Simple but slow convergence
- 2 **Ishikawa Iteration** (2-step): Intermediate complexity
- 3 **Noor Iteration** (3-step): More sophisticated, faster convergence

Historical Context

- **Mann (1953):** Introduced single-step iteration method for finding fixed points
- **Ishikawa (1974):** Extended Mann's method with two intermediate steps
- **Noor (2000):** Further generalization with three intermediate steps, building on earlier work

The progression represents increasing levels of sophistication in handling:

- Nonexpansive mappings
- Contractive mappings
- General monotone operators
- Fixed points in ordered Banach spaces

Key Reference: Pathak's comprehensive treatment in Chapter 5 on Fixed Point Theorems.

Noor Iteration: Basic Definition

Definition (Noor Iteration Scheme)

Let $T : \mathcal{D}(T) \subseteq X \rightarrow X$ be a mapping and $\{a_n\}, \{b_n\}, \{c_n\}$ be sequences in $[0, 1]$.
The Noor iteration is defined by:

$$z_n = (1 - c_n)x_n + c_n T(x_n) \quad (1)$$

$$y_n = (1 - b_n)x_n + b_n T(z_n) \quad (2)$$

$$x_{n+1} = (1 - a_n)x_n + a_n T(y_n) \quad (3)$$

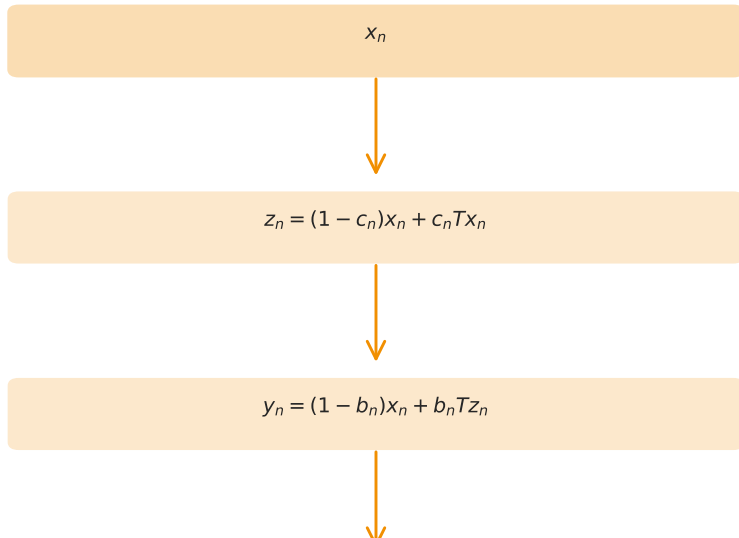
where $x_0 \in \mathcal{D}(T)$ is an arbitrary starting point and $n = 0, 1, 2, \dots$

Key Features:

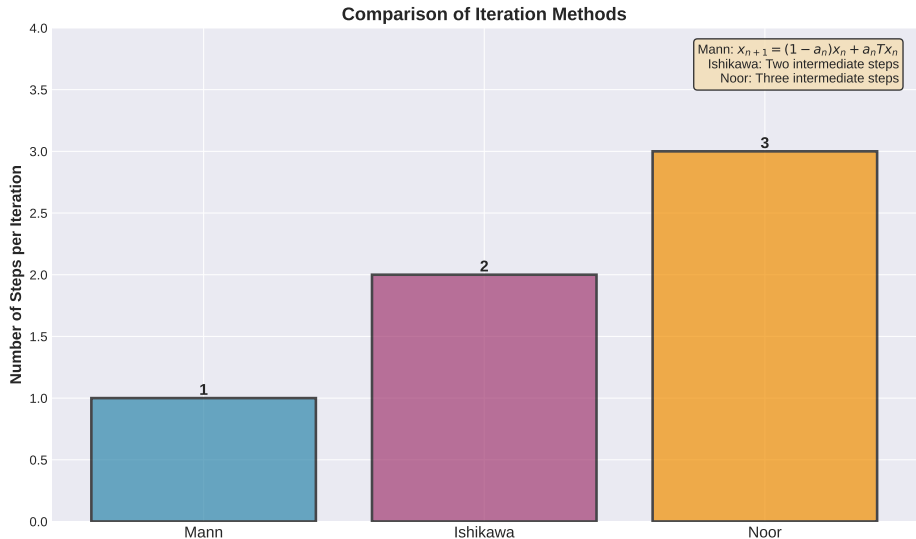
- Three intermediate computations per iteration
- Parameters a_n, b_n, c_n control the convex combinations
- Reduces to Mann when $b_n = c_n = 0$
- Reduces to Ishikawa when $c_n = 0$

Noor Iteration: Pictorial Representation

Noor Iteration: Three-Step Process



Comparison with Mann and Ishikawa



Convergence Requirements for Noor Iteration

Theorem (Convergence Conditions for Noor Iteration)

Let $(X, \|\cdot\|)$ be a Banach space and $T : \mathcal{D}(T) \subseteq X \rightarrow X$ be a mapping.
The Noor iteration $\{x_n\}$ converges strongly to a fixed point $x^* \in \mathcal{D}(T)$ if:

(i) T is a **contraction mapping** with constant $k < 1$, i.e.,

$$\|T(x) - T(y)\| \leq k\|x - y\| \quad \forall x, y \in \mathcal{D}(T)$$

(ii) The parameter sequences satisfy:

- $\sum_{n=0}^{\infty} a_n = \infty$ (infinitude condition)
- $a_n, b_n, c_n \rightarrow 0$ as $n \rightarrow \infty$ (diminishing step sizes)
- $\sum_{n=0}^{\infty} a_n^2 < \infty$ (summability condition)

(iii) $x_0 \in \mathcal{D}(T)$ is an arbitrary starting point

Fixed Point Theorems - Banach Contraction Principle

Theorem (Banach Fixed Point Theorem)

Let (X, d) be a complete metric space and $T : X \rightarrow X$ a contraction mapping with Lipschitz constant $k < 1$.

Then:

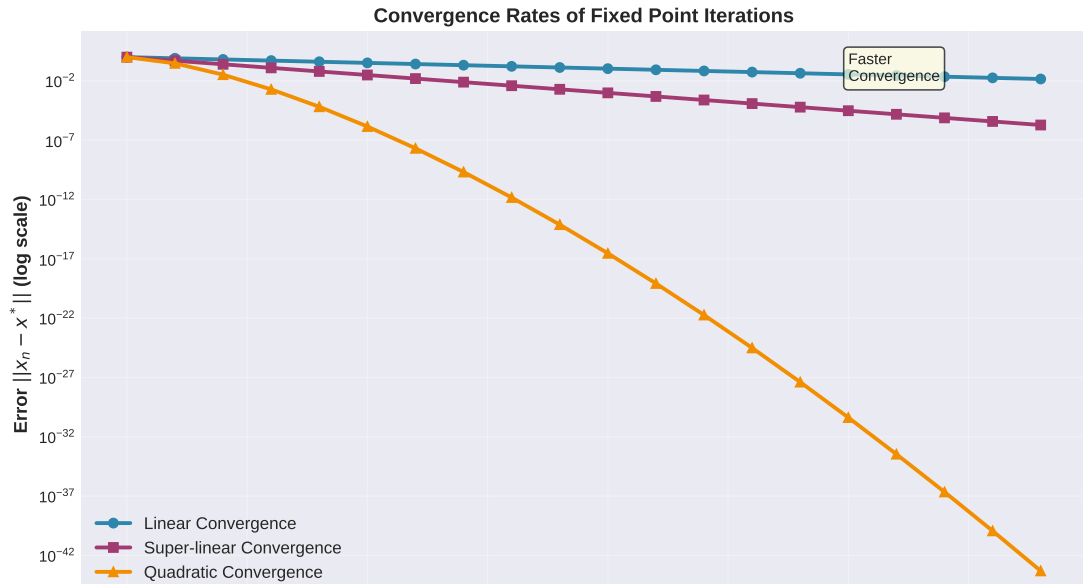
- (i) T has a **unique fixed point** $x^* \in X$
- (ii) For any $x_0 \in X$, the Picard iteration $x_{n+1} = T(x_n)$ converges to x^*
- (iii) The error estimate holds:

$$\|x_n - x^*\| \leq k^n \|x_1 - x_0\|$$

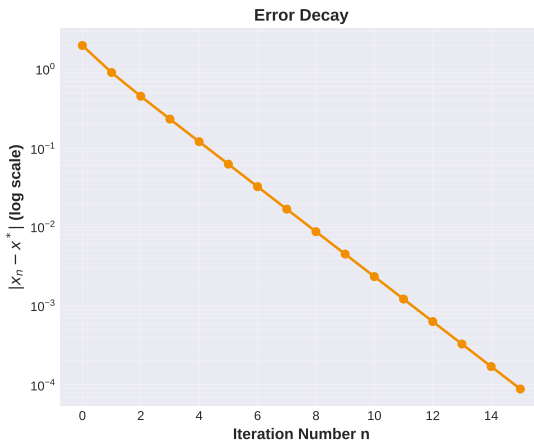
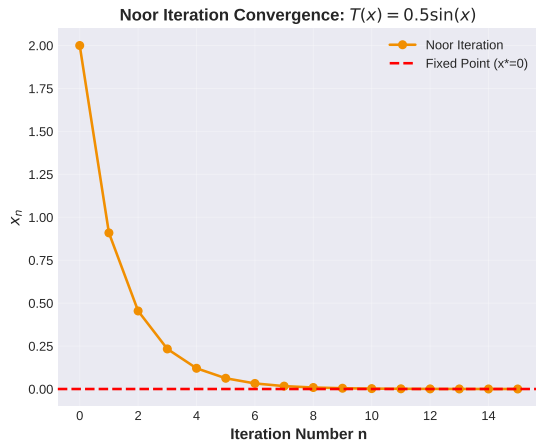
Relevance to Noor Iteration

The Banach principle guarantees existence and uniqueness of fixed points. Noor iteration accelerates convergence by using multi-step schemes.

Convergence Behavior



Example: Numerical Convergence



Setup: $T(x) = 0.5\sin(x)$ with fixed point at $x^* = 0$

- Left: Sequence $\{x_n\}$ converges to fixed point
- Right: Error $|x_n - x^*|$ decays exponentially
- Parameters: $a = 0.7$, $b = 0.6$, $c = 0.5$ (constant sequences)

Ordered Banach Spaces and Cones

Definition (Cone)

A subset K of a Banach space X is called a **cone** if:

- (i) K is closed
- (ii) If $u, v \in K$, then $\alpha u + \beta v \in K$ for all $\alpha, \beta \geq 0$ (convexity)
- (iii) $x, -x \in K \implies x = 0$ (no pair of opposite directions)

Definition (Ordered Banach Space)

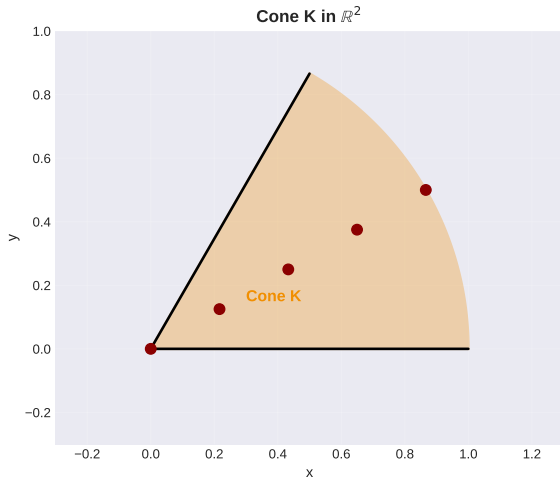
A Banach space X with a cone K is denoted (X, K) and called an **ordered Banach space**.

A partial order relation is defined by: $x \preceq y$ if and only if $y - x \in K$.

Common Examples:

- $(C[a, b], K)$ where $K = \{x \in C[a, b] : x(t) \geq 0 \text{ for all } t \in [a, b]\}$
- (L^p, K) where K consists of non-negative L^p functions

Properties of Cones



Cone Properties (Definition 5.70)

- (i) Closed set
- (ii) Convexity: If $u, v \in K$, then $\alpha u + \beta v \in K$ for all $\alpha, \beta \geq 0$
- (iii) $x, -x \in K \iff x = 0$
(Only origin in both K and $-K$)

Normal Cone (Definition 5.73)

A cone K is normal if $\exists \delta > 0$:

$$\|e_1 + e_2\| \geq \delta, \text{ for } e_1, e_2 \in K, \|e_1\| = \|e_2\| = 1$$

Definition (Normal Cone)

A cone K is **normal** if there exists $\delta > 0$ such that:

Fixed Point Theorems in Ordered Banach Spaces I

Theorem (Krasnoselski Fixed Point Theorem)

Let (X, K) be an ordered Banach space with K a normal cone. Let $u_0, v_0 \in X$ with $u_0 \preceq v_0$, and let $F : [u_0, v_0] \rightarrow X$ be a continuous mapping such that:

- (i) $u_0 \preceq F(u_0)$ and $F(v_0) \preceq v_0$
- (ii) F maps $[u_0, v_0]$ into a compact subset of X

Then F has at least one fixed point in the order interval $[u_0, v_0]$.

Theorem (Isotone Operator Fixed Point Theorem)

Let $T : X \rightarrow X$ be isotone (order-preserving) on the cone K , i.e., $x \preceq y \implies T(x) \preceq T(y)$.

If $T(0) \in K$ and the sequence $\{T^n(0)\}$ has a convergent subsequence in a compact subset, then T has a fixed point in K .

Monotone Operators and Fixed Points

Definition (Monotone Operator)

Let $T : \mathcal{D}(T) \subseteq X \rightarrow X^*$. Then T is **monotone** if:

$$\langle T(x) - T(y), x - y \rangle \geq 0 \quad \forall x, y \in \mathcal{D}(T)$$

Connection to Iteration

- Monotone operators arise naturally in variational problems
- Fixed points of monotone operators can be found via Noor iteration
- The structure of monotone operators ensures convergence under mild conditions

Theorem (Monotone Operator Fixed Point)

If T is a monotone operator on a reflexive Banach space and $0 \in T(X)$, then fixed point iteration methods (including Noor) can be used to find solutions to $T(x) = 0$.

Python Implementation: Basic Setup

```
import numpy as np

def noor_iteration(T, x0, n_iter, a_seq, b_seq, c_seq):
    """
    Perform Noor iteration to find fixed point of T.

    Parameters:
    -----
    T : callable
        The mapping (function) T
    x0 : float or array
        Initial point
    n_iter : int
        Number of iterations
    a_seq, b_seq, c_seq : array-like
        Sequences of parameters {a_n}, {b_n}, {c_n}

    Returns:
    -----
    sequence : array
        Sequence of iterates {x_n}
    """
    x = x0
    sequence = [x]

    for n in range(n_iter):
        a_n = a_seq[n] if n < len(a_seq) else a_seq[-1]
```


Example 1: Finding Square Roots

```
# Find sqrt(2) using Noor iteration
# Fixed point of T(x) = (x + 2/x) / 2

def T(x):
    return (x + 2.0/x) / 2.0

x0 = 2.0
n_iter = 10
a_seq = np.ones(n_iter) * 0.7
b_seq = np.ones(n_iter) * 0.6
c_seq = np.ones(n_iter) * 0.5

sequence = noor_iteration(T, x0, n_iter, a_seq, b_seq, c_seq)

print("Iteration Results:")
for n, x_n in enumerate(sequence[:8]):
    error = abs(x_n - np.sqrt(2))
    print(f"x_{n} = {x_n:.10f}, error = {error:.2e}")
```

Output:

```
x_0 = 2.0000000000, error = 4.14e-01
x_1 = 1.5828427125, error = 1.62e-02
x_2 = 1.4247869564, error = 9.37e-03
x_3 = 1.4144734524, error = 1.64e-04
```

Example 2: Nonlinear Equation

```
# Solve  $x = \cos(x)$  using Noor iteration
# Equivalently, find fixed point of  $T(x) = \cos(x)$ 

def T(x):
    return np.cos(x)

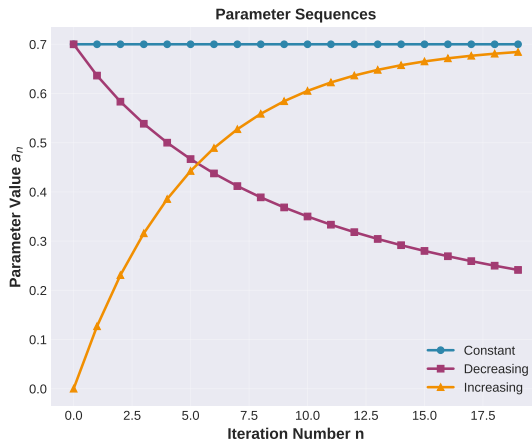
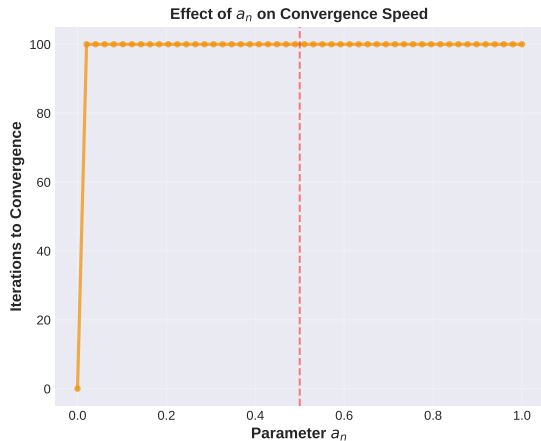
x0 = 0.5
n_iter = 15
a_seq = 0.7 / (1 + 0.1 * np.arange(n_iter)) # Decreasing
b_seq = 0.6 / (1 + 0.1 * np.arange(n_iter))
c_seq = 0.5 / (1 + 0.1 * np.arange(n_iter))

sequence = noor_iteration(T, x0, n_iter, a_seq, b_seq, c_seq)

# Known fixed point (Dottie number)
dottie_number = 0.7390851332 # Fixed point of  $\cos(x)$ 

print("Convergence to  $x = \cos(x)$ :")
print(f"Dottie number (exact): {dottie_number:.10f}")
print(f"x_0 = {sequence[0]:.10f}, error = {abs(sequence[0] - dottie_number):.2e}")
print(f"x_10 = {sequence[10]:.10f}, error = {abs(sequence[10] - dottie_number):.2e}")
print(f"x_15 = {sequence[15]:.10f}, error = {abs(sequence[15] - dottie_number):.2e}")
```

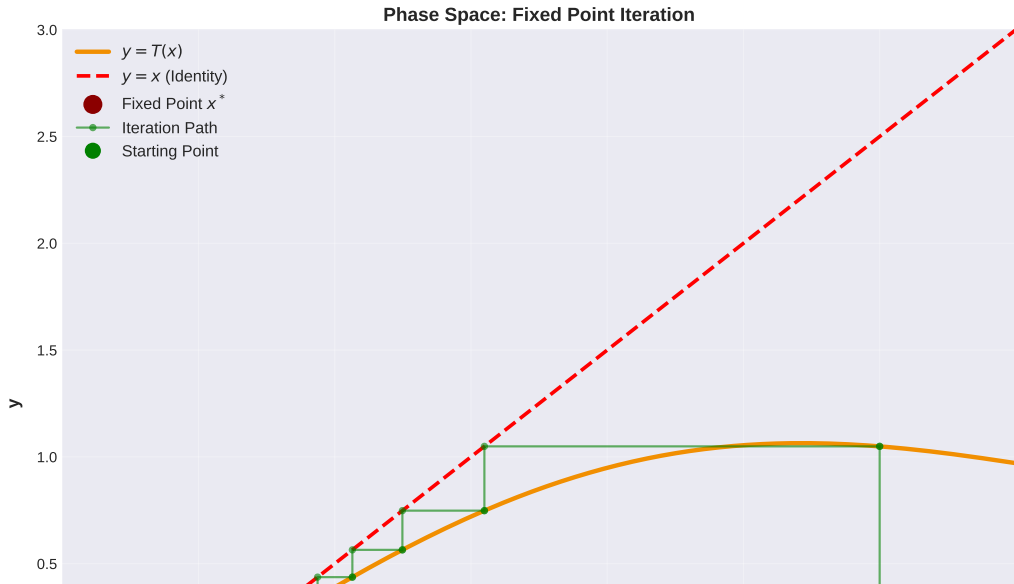
Parameter Selection and Optimization



Parameter Choice Strategies:

- **Constant Parameters:** Simple, $a_n = a$, $b_n = b$, $c_n = c$ for all n
- **Decreasing:** $a_n = a_0/(1 + \beta n)$ ensures convergence conditions
- **Optimal Selection:** Choose parameters to minimize total computational cost

Phase Space Analysis



Convergence Under Weak Contractions

Definition (Weak Contraction)

A mapping $T : X \rightarrow X$ is called a **weak contraction** if:

$$\|T(x) - T(y)\| < \|x - y\| \quad \text{for all } x \neq y$$

without requiring a fixed Lipschitz constant.

Theorem (Edelstein's Theorem)

If T is a weak contraction on a complete metric space X , and the orbit $\{T^n(x_0)\}$ has a convergent subsequence, then the limit point is a unique fixed point of T in a neighborhood of x_0 .

Application to Noor: Noor iteration converges even when T is only a weak contraction, provided parameter sequences satisfy convergence conditions.

Nonexpansive Mappings

Definition (Nonexpansive Mapping)

A mapping $T : C \rightarrow C$ is **nonexpansive** if:

$$\|T(x) - T(y)\| \leq \|x - y\| \quad \forall x, y \in C$$

Comparison with Contractions

- Nonexpansive maps need not have fixed points (unlike contractions)
- Require additional conditions: C compact or convex
- Multiple fixed points are possible
- Noor iteration handles nonexpansive mappings with appropriate modifications

Theorem

For nonexpansive mappings on compact convex sets in Hilbert spaces, Noor iteration with appropriate parameter sequences converges weakly to a fixed point.

Weak Convergence vs Strong Convergence

Definition (Strong Convergence)

A sequence $\{x_n\}$ in a Banach space X is **strongly convergent** to $x^* \in X$ if:

$$\lim_{n \rightarrow \infty} \|x_n - x^*\| = 0$$

Definition (Weak Convergence)

A sequence $\{x_n\}$ in X is **weakly convergent** to $x^* \in X$ if:

$$\lim_{n \rightarrow \infty} \langle f(x_n) \rangle = \langle f(x^*) \rangle \quad \text{for all } f \in X^*$$

For Noor Iteration:

- Guarantees **strong convergence** for contractions
- Achieves **weak convergence** for nonexpansive mappings in certain spaces
- Choice of Banach space structure affects convergence type

Stability and Perturbation Analysis

Stability Question

If T is slightly perturbed to $T_\epsilon = T + \epsilon F$, do the fixed points vary continuously?

Theorem (Implicit Function Theorem Application)

If T is continuously Fréchet differentiable and $\mathbf{1} - DF(x^)$ is invertible (where x^* is the fixed point), then fixed points vary continuously with small perturbations.*

For Noor Iteration:

- Stability of the iteration depends on the spectral radius of $DF(x^*)$
- Smaller step sizes (a_n, b_n, c_n) increase stability
- Robustness to rounding errors increases with smaller parameters

Application 1: Differential Equations

Example (Two-Point Boundary Value Problem)

Consider the boundary value problem:

$$\begin{aligned} -x''(t) + h(t)x(t) &= f(t), \quad t \in (0, 1) \\ x(0) &= x(1) = 0 \end{aligned}$$

This can be rewritten as a fixed point problem: $x = Gx$ where

$$G(x)(t) = \int_0^1 G_0(t, s)[f(s) + h(s)x(s)]ds$$

with G_0 the Green's function.

Solution Strategy:

- 1 Show G is a contraction (or weak contraction) on $C[0, 1]$
- 2 Apply Noor iteration to compute the solution
- 3 Use parameter sequences to ensure rapid convergence

Application 2: Integral Equations

Example (Volterra Integral Equation)

The second-kind Volterra equation:

$$x(t) = f(t) + \int_0^t K(t, s, x(s)) ds$$

defines a fixed point problem where $T(x)(t) = f(t) + \int_0^t K(t, s, x(s)) ds$.

Analysis:

- Contractivity depends on kernel K and bounds
- Noor iteration provides numerical solution with error control
- Parameter sequences optimize computational efficiency
- Applications: heat equations, diffusion, kinetic equations

Theorem (Convergence for Volterra Equations)

If $|K(t, s, x_1) - K(t, s, x_2)| \leq L|x_1 - x_2|$ with small enough Lipschitz constant L , Noor iteration converges to the unique solution.

Application 3: Optimization Problems

Minimization via Fixed Points

Optimization problems can be converted to fixed point problems:

$$\min_{x \in \mathcal{C}} f(x) \quad \Rightarrow \quad x = P_{\mathcal{C}}(\nabla f(x))$$

where $P_{\mathcal{C}}$ is the projection onto the convex set $\mathcal{C}$.

- Gradient descent is a special case of iteration
- Proximal methods use fixed points of proximal operators
- Nesterov iteration can accelerate convergence in optimization
- Important in machine learning and signal processing

Example

Proximal gradient algorithm for minimizing $f(x) + g(x)$:

$$x_{n+1} = \text{prox}_g(x_n - \nabla f(x_n))$$

Key Takeaways

- 1 **Noor Iteration** is a three-step fixed point iteration method that generalizes Mann and Ishikawa iterations
- 2 **Convergence** is guaranteed under:
 - Contractive mappings (or weak contractions)
 - Appropriate parameter sequences satisfying theoretical conditions
 - Proper choice of Banach space structure
- 3 **Fixed Point Theorems** (Banach, Krasnoselski, etc.) provide the theoretical foundation for guaranteeing existence and uniqueness of fixed points
- 4 **Applications** span differential equations, integral equations, optimization, and numerical analysis
- 5 **Practical Implementation** requires careful parameter selection to balance convergence speed with computational cost

Further Reading

-  Pathak, H.K. (2018) *An Introduction to Nonlinear Analysis and Fixed Point Theory* Springer
-  Kirk, W.A. (2001) Fixed point theorems in nonempty compact convex sets *In: Handbook of Metric Fixed Point Theory*
-  Browder, F.E. (1965) Nonexpansive nonlinear operators in Banach spaces *Proceedings of the National Academy of Sciences*
-  Mann, W.R. (1953) Mean value methods in iteration *Proceedings of the American Mathematical Society*, 4(3), 506-510
-  Ishikawa, S. (1974) Fixed points and iteration of commuting nonexpansive mappings *Pacific Journal of Mathematics*

Conclusion

Significance of Noor Iteration

Noor iteration represents a sophisticated yet practical approach to finding fixed points in nonlinear functional analysis. It combines:

- Mathematical rigor through fixed point theorems
- Computational efficiency through multi-step schemes
- Practical applicability to real-world problems

Future Directions

- Acceleration techniques (Anderson, Nesterov-type)
- Parallel and distributed implementations
- Applications to machine learning and deep learning
- Variable order step methods

The study of fixed point iterations continues to be central to numerical analysis and